10m Reference List

General Purpose Use

Best Op times are Spring-Summer & Winter **Es** Season.

F-Prop Cycle 25 Start-up **Fall 2020.** Predicted Peak **July 2025.**

2023 New Peak Prediction> **"Dec 2024"**

Dial:	Mode:	Notes:	Notes/Programs:
28.0-28.07+	**CW**	Common activity **< .04**	"**CW Get**" decoder- UA9OV
28.06	**CW**	Legacy QRP Calling Channel	& **SOTA** freq ~28.062- .066
https://sotawatch.sota.org.uk/en/			**(POTA**~ random cw & phone freqs)
28.063+-	**FeldHell**	Hellscreiber/"FM105" for dx, qsb	**IZ8BLY**, FLDigi, MultiPsk
https://sites.google.com/site/feldhellclub/Home/feld-hell-faq			<< FAQs. (*n6spp FH #204*)
28.0675	**CW**	**W1AW** Code Practice M-F	2000z, 2300z, 0200z
28.0675	**CW**	**W1AW** Bulletin M-F	2100z, 2400z, 0300z
28.07-28.12+	**RTTY/Digital**	(Most RTTY 28.08-28.095+-)	FLDigi, HRD, MultiPsk etc
28.07	**Opera**	Weak Signal- Auto Qso & Bcn Mode	EA5HVK (Ok in ITU Region2)
28.074	**FT-8**	Initial 2017 use: Multi-Hop Es/Fast QSB	13.5sec TX seq **(snr to -21 dB)**
https://hamspots.net/alldigi/		<<Select 10m etc	(WSJT-X)
28.076	**JT65-A**	177.6 Hz width per Stn.	Still in use in 2025! **(snr to -25 dB)**
28.078	**JT9-A**	15.6 Hz width per Stn.	+Popular on the **474KHz** WSPR Ch. **(snr to -27 dB)** (WSJT-X)
28.078	**JS8-Call**	Includes- Free message & All Call	See op Notes p.8 **(snr to -24 dB)**
28.095	**Digi Msg**	**W1AW**: PSK31, RTTY, MFSK16	2200z, 0100z Bulletin
28.105	**VarAC**	KB to KB VARA & Beaconing	New 2021. by 4Z1AC
28.1150	**VARA HF**	(to ~28.1350 MHz) ~WinLink Express	+ HF VARA by EA5HVK
28.1188	**Prop Net**	"PNP" **PSK31** Path Reporting System	Multi-window PSK31 TX/RX robot
decoder	https://propnet.org/index3faq.shtml	<<FAQs	***(2005- Dec2018)***
28.12015	**PSK31**	Region 2 Channel	Still in Use- All Regions
28.12015	**PSK31 BCN**	"YV4DHS/B" **AF: nr 1888 Hz**	Venezuela. TX's every 5-10 mins
28.122+-	**Olivia**	8/250 etc. Contest activity 2023	See HamSpots.net

10m Reference List cont.

Dial:	Mode:	Notes:	Notes/Programs:
28.1246	**WSPR**	Weak Signal Prop Reporter	"Whisper"(WSJT-X) **(snr to -31 dB)**

http://www.wsprnet.org/drupal/wsprnet/map
One-way Prop mode. **5.9 Hz width** per stn! Look for thin vertical lines in your waterfall @ 1400-1600 Hz **AF**.
Note-There is a **REAL 2-Way WSPR QSO mode**. It's located in "WSJT-v.7" **only**. JT9-1&2 replaced it **circa 2010**.
N6SPP: *"At 5.9Hz width- loading a window frame with a counterpoise wire along the carpet, will net DX spots"*

28.1246 **QRSS/FSK CW** **Super Slow CW Rcvr Test** (Radio in USB mode!) ITU Reg.1, 2, 3 **Use "ARGO". Select Horiz Sweep & Normal speed. Palette: lighter. AGC: low. QRSS3 mode.** Look for **VE1VDM** & other stns **+- 980-1200 Hz AF**. **Read (the TOP) long dits & dahs** from left to right in the waterfall. (info: QRSS Knights @ groups.io) Popular just below the 20, 30 & 40m WSPR channels too.

28.13 +-	**FSQ**	"Fast Simple QSO" mode Image & Keyboard to Keyboard	Proposed freq-Region2 ? **FLDigi**
28.14	**ROS**	Robust KB to KB, multi-baud (Used in the U.S. in **2009, 2010**)	~EA5HVK/**Region 1 & 3 Only** (**Reg.1&3**: 27.635 MHz, prop)
28.145	**ISCAT-A**	Legacy Iono-Scatter (good for 300+ miles) (N.AM Activity replaced by FT-8 **~2017**)	**OK in Region 2** (WSJT-X) **(snr to -17)** Try with **Es** !
28.145	**JT9-G**	Hi-Speed-Meteor-Scatter/**HSMS**	**890.6** Hz width per stn (snr -17 dB)

JT9-G N6SPP **How-To** Link>> http://tinyurl.com/3jcycb3w Ok in Region 2
(28.148 is the default. 28.145 is used in all Regions to avoid the IW0CPK/B on 28.1475 MHz)

28.145+-	**MSK-144**	<< HSMS- BW **up to 2800 Hz** allowed	**New 10-160m FCC rule ~ Dec.2023**
28.145+-	**Q65**	<< Common op: 433 Hz wide, 15sec seqs (snr: **-23 to -37**dB, width dependent)	See PingJockey Central (WSJT-X)
28.146	**ALE**	Auto Link Establishment/Bcning,Tfc	HFLINK Group 2019 (prog: **MultiPsk**)
28.18	**FT-4**	General use & contesting.	In the **DX Bcn subband**/Freq change?
28.2	**CW**	**IARU/NCDXF International Bcn Network**	Bcn Ntwork Ch: 14.1, 18.11, 21.15,

24.93, 28.2 MHz. There are about 16 (of 18) Active beacons here. Listen for "**W6WX**"in grid:CM97
https://www.ncdxf.org/beacon/ **<<< Live Status** Mobile app: **BeaconAid-HF**

IARU Beacon Map Follow the IARU beacons on a small map. By PA1ARE
https://www.qsl.net/pa1are/software.html **(Old, but works ok with Windows 10,11)** Note-KH6WO is now KH6RS. Ensure pc clock is accurate.
HamClock & many logging programs include an IARU Beacon tracking map.

28.2-28.3 CW **USA & DX Beacon Subband** +Other DX Bcns down to 28.15+-MHz
https://storage.iaru-r2.org/index.php/s/mDCqpdaf8AXt3An <<Region 2 IARU 10m Beacon list
(N.AM Bcn coordinator: WJ50> WI5V> now AC5JM)

10m Reference List cont.

Common SFO Bay Area Beacon Catches:

	CW id:	QTH/Grid:	Local+DX **Reports/Notes:**
28.166-.168	XE2O/B	nr Monterrey/DL95ur	2024 Fall (+ 50.068 MHz)
28.168	VA3KAH/B	E.Lake Huron/EN95wc (IOTA)	2024 Fall (**see QRZ**)
28.1753	VE3TEN/B	Ottawa Club/FN25ei	2023 Winter(tone **280**Hz,id **660**Hz)
28.193	VE4ARM/B	S.Cent Manitoba/EN09mw	2024 Fall
28.197	VE7MTY/B	SE of Vancouver/CN89pf	Regular Catch! **Go-to Bcn#1**
28.2	**W6WX**	Mt.Umunhum/CM97ae	2025 **Local IARU SFO area Beacon**
28.204	W0WF/B	NW of St.Louis/EM48rr	2024 Fall
28.209	N5TIT/B	Dallas,TX/EM12px	2024 Spring
28.2093	K7EMX/B	SW of Salt Lake City/DN30xp	2024 Fall (+50.0625)
28.212	AC7GZ/B	SE of Phoenix/DM43bh	2025 Jan.
28.2124	LU7DQP/B	Buenos Aires/GF05th	2022 Summer (watch)
28.2143	W6REK/B	**San Jose**/CM97ah	2024 Fall//Henry's
28.2178	K4PAR/B	C.Georgia/EM83ii	2024 Fall
28.2179	**N6SPP/B**	Pittsburg,CA/CM98ba	2018-Present,new xmtr qth
28.2279	**ZL3TEN/B**	N.Zealand/RE66rj	2025 Jan. (6/2024@2247z,cm98)
28.2288	N6RVI/B	San Diego/DM12mr	2025 Feb.
28.23	W7PFR/B	E.of Eatonville, WA/CN86vt	2023 Fall
28.2308	NG7I/B	E.Cent WA/DN07tc	2025 Jan.
28.2365	W0KIZ/B	S. of Denver/DM79nl	2024 Spring
28.244	WA6APQ/B	Los Angeles/DM03wt	2024 Fall
28.259	AI7AJ/B	Phoenix/DM33xk	2024 Spring
28.2651	K6SSB/B	E.of **Groveland**, CA/DM07as	2025 Feb.
28.2705	PY4MAB/B	Brazil//GG68re	2025 Jan.
28.2785	LU7DW/B	Buenos Aires/GF05sh	2024 Summer (by CT2 & W5)
28.28	**K5AB/B**	Austin/EM01rk	2025 Jan. **Go-to #2**
28.285	W7IEW/B	Olympia/CN87na	2023 Summer
28.2923	**NH6HI/B**	Kauai/BL01fw	2023 Summer(NH6HI<>N6SPP spots)
28.2962	KA7BGR/B	Medford, OR/CN82nk	2024 Summer
28.298	K7PO/B	Phoenix/DM33nl	2022 Spring
28.3	K6FRC/B	nr **Tracy**, CA/CM97hp	2025 Feb. /Paul's (+**50.08MHz**)

Tip: Run "**ARGO**" to watch cw/digi sigs in a simple waterfall @ different speeds. **For Beacons**- Select Horizontal sweep (RT Click). Use CW/NDB mode & Normal Speed. Radio in CW mode. **Beacons** will appear near 550-700 Hz AF.

Note: I'm looking for **Active** Brazil & Chile Beacons to add here. Please email me if you have info. TU!

Look For: **ZB2TEN/B Gibraltar 28.17 MHz** in IM76hd. Reports: Summer 2024 by CT2, F6, VE3. Fall 2023 by W2, W3.

10m Reference List cont.

Dial:	Mode:	Notes:	Notes/Programs:
28.327	**SSB**	Beacon Operator's Calling CH. ~Legacy	Call at the Top of each Hour
28.3275	**SSB**	"HF Pack" Channel (now on groups.io)	Backpack, bicycle, portables
28.38	**SSB**	**10-10 Net**, Mon-Sat @ 1800z/& 28.8 MHz	Relays in each U.S. Time Zone
28.385	**SSB**	QRP Calling Channel	
28.4	**SSB**	"Middle of the Tech/General Phone Band"	Hot Spot
28.425	**SSB**	"**KC4TVZ**" Semi-attended CQ Calling Loop Station.	**SSB Prop Beacon**, hi
28.59	**SSB**	**W1AW Voice Bulletin/Active 2025**	**Tues-Sat @ *0145 Zulu***
28.665	**SSB**	OMISS Net. Sat/Sun 1800z (+- 7 KHz)	(n6spp #12218)
28.68	**SSTV**	USA: Scottie 1, 2, DX modes//JA's: Martin1, DX **"mmSSTV"** by JE3HHT	
28.885	**SSB**	**Old, pre-internet 6 Mtr Watch-Liaison Channel!**	
29.01-29.03	**AM**	QSB & Phase conditions on signals.	Try switching ant polarity!
29.25	**FM(try SSB)**	**"C4FM"** mode**.** Look for bcns/sigs nr 2450Hz AF in waterfall. Simplex Calling Freq	
29.3- 29.51	**SSB/CW**	**Satellite Subband**	"AO-7" Mode A/ "NO-84"psk
29.6	**FM**	Simplex Primary	**Listen for receiver desensing**!

REPEATERS

29.62, 29.64, 29.66, 29.68 << Repeater Outputs/ Standard Splits

Check the ARRL Repeater Directory (many entries are outdated), and RepeaterBook.com

Below are a few repeaters worked Direct from the SFO Area:

KQ2H/r	**29.62 out/29.52 in, pl 146.2**	Manhattan, NY	2023 Spring
WB2JQK/r	**29.68 out/29.58 in. pl 107.1**	nr Buffalo, NY	2023 Spring
KX4I/r	**29.64 out/29.54 in, no pl**	Tuscaloosa, AL	2022 Summer
W1OJ/r	**29.62 out/29.52 in, pl 100**	Boston (in-active)	90's/2000's
KR7IS/r	**29.68 out/29.58 in, pl 162.2**	nr Portland, OR	2022 Summer
W6SRR/r	**29.68 out/29.58 in, pl 94.8**	(in-op pending new QTH/Pleasanton, CA area/2021)	
N6TKW/r	**29.68 out/29.58 in, pl 114.8**	Napa, CA (Mt.Veeder) Linked to 51.72out/51.22in,pl 114.8	
N6LIZ/r	**29.66 out/29.56 in, pl 107.2**	Pasadena/Mt.Harvard/grid: DM04xf 2022	
W7RNB/r	**29.68 out/29.58 in, no pl**	N.of Seattle, WA/grid: CN87ts	
W5DFW/r	**29.66 out/29.56 in, no pl**	Dallas, TX	2023 Summer
VE3MMX/r	**29.64 out/29.54 in, no pl**	SW of Toronto/grid: EN93ud	Wk'd **DM3VOT+DO1KRT** thru

this repeater on 8Oct2022 & **PE1OUW, NP4AF, DH1KJ, & ON4BW** on 30Sept2023.

10m Reference List cont.

REPEATERS cont.

KE4QOX/r	**29.68 out/29.58 in, pl 88.5**	S.WV Linked to WW4DC/r N.C & KE4QOX /r (on 29.66) 2023	
K1FB/r	**29.66 out/29.56 in, pl 67**	Nashville, TN	Fall 2023
K9HGX/r	**29.62 out/29.52 in, pl 103.5**	Decatur, IL	Fall 2023
KB3AWQ/r	**29.66 out/29.62 in, pl 173.8**	Williams Port, PA	Fall 2024
KD0WPK/r	**"29.67 out/29.57 in" pl 103.5**	"Iowa 10m Group"	20Oct2024: Sun/Th@1730PDT

10M Repeaters in New Mexico? Colorado? Arizona? Utah? Wyoming?

N6SPP **10M Repeater Map !** ~ SFO Area Paths:
https://drive.google.com/file/d/1Z58V_xGr8aOfQHw1x_z7YjeEAjfR8alS/view?usp=sharing (Zoom in a little)

How To Work 10m Repeaters-

(I'll use KR7IS/r as an example. See Split/PL info above)

During a Regional Es or F prop path opening-

PTT on the KR7IS/r input for 4 seconds, and Un-key for 4 seconds. Repeat until you hear the repeater kerchunk or id.

Next, announce **only your call** (once or twice), and Un-key! Repeat…

The opening to the Target repeater might be available for only a few seconds. If the path to the repeater remains open for more than a few seconds, repeat your call again a few times, and announce your four digit grid, or nearest BIG CITY.

When a station comes back to you:

1. **Exchange Calls**. Make sure you have each other's call **correct** before exchanging any other info. (This is old-school Tip #1. This is **not PSK31**, where it's common to see stations exchange the **"Kitchen Sink info"** before confirming call signs, under poor path conditions. hi)

2. Tell the station you are **working thru** the "KR7IS repeater in Portland, Oregon" -if you have time.

(He might have a beam antenna, and he'll want to maximize his signal into the Portland area- **if** he is working KR7IS direct- using pl 162.2).

Note- It is very common for two or three repeaters that share the same PL and Pair, to **open in tandem** during a good opening. This happens with **non-ctcss repeaters too**.

3. Exchange Grids and/or Signal reports (optional).

Note: Try other repeater pairs and pl's. Store multiple pairs/pl's in your radio's memory. Always listen for **receiver desensing** on all four repeater outputs, and 29.6 FM Simplex. Try the same pair/pl with a vertical and horizontal antenna. Phase & condx are always changing. Check the Repeater Directories for odd splits too.

FM Simplex & Repeater Trick: PTT & call just as the SNR starts to increase. The time (gap) between signal fade and SNR increase is often ~ 5 to 15 seconds.

Common 10m Target or "Pipeline" Paths to/from the SFO Bay Area during previous Solar Cycles:

Roughly~ San Diego, Seattle, Vancouver, Phoenix, Albuquerque, Medford, Billings, Boise, Casper, Denver, and Northern Mexico.
Further out Common Paths to: Dallas, Birmingham, Atlanta, Mid-West, and the Caribbean.
Super Common DX Paths: Argentina, Chile, Brazil (+ other S.AM)
Other Common DX Paths: Oceania (ZL, VK, KH6, FO, T32)
Asia: JA, DU, BA, HL
Asiatic RU: Sakhalin Island "UA0 Freddy Freddy"
Semi-common Paths: Alaska, W. Africa coast (Canary Isles/Madeira, and South to W.Cent AF)

Tips:

SFO Region> Sporadic E- 6 & 10m: Watch for paths to form first- Easterly from San Diego, and Easterly/S.Easterly from Vancouver. And, from S.Diego to/from Vancouver.
An SFO Bay Area path usually follows next to the PAC NW, W0, W5, or LAX. Conditions permitting.

~ **Typical Es Season** Opening Window: 1100-1300 + - **(local),** and often before sunset.
~ Cycle 25 **F Prop** Upswing+ Peak Openings: local daylight and into the late evening
~ **10m- A few past Cycles: Open 24hrs/Day !**

If you work a station on 28.xxx SSB and condx are good, have them qsy to 29.6 FM, and then 29.01 AM for a try. **"Three modes with one station, in three minutes".**

If a stn is **> - 8** dB on **FT-8** (28.074 MHz), after sending 73, put "**Call Me 28.33**" in the **Free Message Box** and Send 3 times. QSY + call them on 28.33 SSB and try for a phone QSO. **This old JT65A trick works!**

Listen for 11Mtr activity on **27.385 LSB, 27.025 AM, and 27.165 AM.** This is a great propagation tool! There is always someone, somewhere transmitting on these frequencies. Good indicator of DX and Stateside openings.

10m Reference List cont

Listen For the steady tones of FT-8 on 28.074 MHz for openings.

Look for the very thin, 5.9 Hz, vertical traces of **WSPR** stations on **28.1246 MHz USB**. Signals will appear between **1400 to 1600 Hz AF in a standard waterfall.**

Check for **CW activity** between 28.01 and 28.05 MHz +-.
Tune slowly in the Beacon subband. Try with a wide filter too.

Don't forget about 12m! If 10 and 11 are open, 12 is likely open too.

Links/Tools:

Sporadic E **Video/Info** https://3fs.net.au/sporadic-e-propagation/ By **VK3FS** circa 2020

DxMaps.com
https://www.dxmaps.com/spots/mapg.php?Lan=E&Frec=28&ML=M&Map=NA&HF=N&DXC=ING2&GL=N
LF-UHF Path Plots. See 28MHz tab. **Try the "15 minute" Filter!** Formerly called "DX Sherlock".

HamSpots.net Great Digital/WSJT Tool
See where your digital signal has been auto-spotted in the last several minutes. *Modern & accurate.*

PskReporter.info https://pskreporter.info/pskmap.html Are you being spotted?! >Enter Call + Mode.
Or, search for DX + Cluster Spots.
Note- This is a 20+ year old platform. Cross-ref Spots with a DX Cluster for mode & freq accuracy if necessary.

DxHeat.com (dx cluster) https://dxheat.com/dxc/ ***Nice & New.*** Look for 28MHz Reports of activity in your Region.

ON4KST 28/50 MHz LIVE **Popular with 50 MHz Ops. Click the "50 MHz IARU Region 2 Chat" link. This is where the 6m Heavy Hitters Chat & Sked.** http://www.on4kst.com/chat/start.php (Site contains VLF-SHF maps & chat)

DxSummit.fi (dx cluster) http://www.dxsummit.fi/#/?include=28MHz **Easy Callsign Activity Search !**

Sporadic E- N.Am Map https://www.tvcomm.co.uk/g7izu/radio-propagation-maps/north-american-sporadic-e/
Near-live **28 to 144 MHz** Path Lines + MUF values.

Max Usable Freq Map https://prop.kc2g.com/ Popular & simple. Look for >24MHz values

10m Reference List cont.

Reverse Bcn Network https://beta.reversebeacon.net/main.php?zoom=41.01,-192.01,2.50&rows=100&spotters=1&max_age=2,days&bands=10&spotted_call=&hide=distance_km
All bands.. Give it a try! In CW mode- Call CQ de Your Call three times. Select Band & look for your Call in the list. Repeat your CQ & check the list again to see if you've been spotted by a 'skimmer' (**cw robot rcvr**).
Also good for searching beacon callsigns on 6 & 10M.

Grid Map https://www.k7fry.com/grid/?qth=CM87ST98
10 Character !! Plot Distance and Bearing between two locations. Lat/Long too. (Google imagery)

WSJT-X on FB https://www.facebook.com/groups/WSJTX

WSJT-X Latest Version https://wsjt.sourceforge.io/wsjtx.html

JS8 Call (by KN4CRD) https://www.facebook.com/groups/JS8CALL KB to KB, with nice beaconing mode.
JS8 Call note- Messages are truncated/ limited to 13 characters like JT/FT mode transmissions! **No RX/TX timing sync between stations.** An "End of Msg" symbol is displayed@ the end of each transmission. Normal to Turbo modes.

Path Predictions https://www.voacap.com/hf/ Set **Red Home QTH** marker/Drag **Blue DX** marker, set mode, output pwr, and approx antenna for each band. Click **"PROP WHEEL".** Hover curser around the Wheel for Band & Zulu Path time.

Other Links-

N6SPP's FB Group https://www.facebook.com/groups/596720540352567/ **"HF Radio Plus"**

BCB FREQ- Quick Search http://www.mwlist.org/mwlist_quick_and_easy.php?area=3&kHz=740
"AM" Radio

Tropo-Ducting Map Site **https://vhf.dxview.org/#** Zoom in and click on a callsign to see path details.(Paths: via 144MHz APRS)

630/2200 meters & MedFer Band https://www.lwca.net/mb/index.html
Antennas, reports and more. Also check the "**630/2200m"** tab at DxMaps.com

Other Freqs-

40 MHz/ 8 Meter Band:

40.68 MHz USB- N.AM Experimental License Stns~ WSPR/FT-8 Dial freq. 2021-present.
S.Africa and many EU countries have had transmitting privileges here for years.
See EI7GL's site/blog for up to date info and experimental info.

40 MHz N.America Experimental Stns Map !
Mode: FT8 (+ WSPR, Q65, CW, SSB)

https://drive.google.com/file/d/1tA_jux5YdAd4obbIAZfqbnJVZEvHbhbX/view?usp=drive_link

Note: Fall 2023> Possible change to USA licensing. New Stn in 2023> ZF1DM ~Caymans.
VE6PG/VE3IIM waiting for new 8m license ~ Jan.2024 (map: n6spp)

Sporadic E/Meteor receive test freq:

40.6692+ - MHz USB- "SNOTEL" Snow Level scada transmissions 24/7 via meteors (+ Sporadic E)

(Lots of sigs here during the Spring/Summer & Winter Es Season)

28.675 + 28.69 MHz USB- XE1SDZ SSTV Scottie 1,2. Grid EK09kk, Mexico City. Summer 2022

11m:

27.535 MHz LSB? Repeater- Costa Rica 2022
27.700 MHz USB ~Pirate~ SSTV- Scottie 1, Scottie DX (mmSSTV program) 2022- Present
27.245 MHz USB JS8-Call
27.265 MHz USB FT-8

For a "28-432 MHz Ref", "60m Ref", & "8m Ref" Quick List, pls see the links on my QRZ page.
73, n6spp 10-10: 77685